

Stellar Trace: Host Profiling & Machine Learning

Session 1: Observational Selection & Non-Parametric Statistics (Tasks 7–9)

Astronomy Club

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Phase 2 Structure

Stellar Trace Phase 2 transitions from the volume-complete universe model of Milestone 1 to real observational data ingestion, classifier training, and physical model fitting:

Session 1 (Today): Telescopes & Statistics (Tasks 7–9)

- Task 7: Observational Selection Effects & Malmquist Bias.
- Task 8: Ingestion, Coordinate Cross-Matching, and SQL WAF Bypass.
- Task 9: 1D vs. 2D Kolmogorov-Smirnov statistical testing.

Session 2: Machine Learning & Data Leakage (Tasks 10–12)

- Oversampling under strict train-test separation; MLP Classifiers.

Session 3: Rate Fitting & Physics-Informed nets (Tasks 13–15)

- Bayesian MCMC, PINNs, and DTD Convolution.

- In **Milestone 1**, you generated a volume-complete synthetic universe based on galaxy mass functions and star-forming correlations.
- **The Problem:** Real telescope catalogs are not volume-complete. They are limited by the telescope's mirror size, detector sensitivity, and atmospheric dust.

Today's Goals:

1. Mathematically apply telescope sensitivity thresholds (**Malmquist Bias**) to our mock universe.
2. Ingest real Core-Collapse and Type Ia supernova host catalog data.
3. Write coordinate cross-matching algorithms to link supernovae to their hosts.
4. Query the SDSS spectroscopic database server and bypass firewall restrictions.
5. Run 1D and 2D non-parametric statistics to prove that supernovae live in distinct environments.

Section 1: Observational Selection & Malmquist Bias (Task 7)

Volume-Complete vs. Flux-Limited Surveys

To understand real telescope data, we must differentiate between two types of astronomical surveys:

- **Volume-Complete Survey:** Contains every single galaxy within a specified physical volume V (comoving sphere), regardless of how bright they appear. Extremely difficult to achieve because small, faint dwarf galaxies are missed.
- **Flux-Limited Survey:** Only includes objects whose flux on the sky is higher than the telescope's detector sensitivity threshold:

$$f > f_{\text{limit}}$$

- **SDSS Spectroscopic Limit:** The Sloan Digital Sky Survey spectroscopic catalog is complete down to a limiting apparent magnitude of $m_r < 23.5$ in the r -band. Any galaxy fainter than this is ignored.

Light Propagation

The flux of light received on Earth scales inversely with the square of the luminosity distance D_L :

$$f = \frac{L}{4\pi D_L^2}$$

- For a fixed flux limit f_{limit} , the minimum detectable luminosity increases quadratically with distance:

$$L_{\text{min}}(z) = 4\pi D_L(z)^2 f_{\text{limit}}$$

- At low redshifts (close to us), we can detect both massive, bright galaxies and tiny, faint dwarf galaxies.
- At high redshifts (far away), dwarf galaxies fall below the detection limit and disappear.
- **Malmquist Bias:** The systematic selection of high-luminosity objects at large distances. If uncorrected, we would falsely conclude that distant galaxies are systematically more massive.

Mathematical Model Derivation: Step 1 & 2

To apply the survey limit ($m_r < 23.5$) to our mock galaxies, we convert their simulated properties (M_* , sSFR, z) into r -band apparent magnitude (m_r):

1. **Step 1: Specific Star Formation Rate (sSFR) to rest-frame ($g - r$) color:**
Young, massive stars are hot and blue; old, low-mass stars are cool and red:

$$(g - r) = \text{clip}(0.5 - 0.25 \cdot (\text{sSFR} + 10), -0.1, 1.0) + \eta$$

where $\eta \sim \mathcal{N}(0, 0.08)$ represents intrinsic astrophysical scatter.

2. **Step 2: Bell & de Jong (2001) Mass-to-Light (M/L) ratio in g -band:** Stellar population synthesis models relate color directly to M/L in the g -band:

$$\log_{10}(M/L)_g = 1.5 \cdot (g - r) - 0.7$$

Translating M/L from g -band to r -band

How do we find the Mass-to-Light ratio in the r -band, $(M/L)_r$?

- The rest-frame color $(g - r)$ is defined as the difference in absolute magnitudes:

$$(g - r) = M_g - M_r$$

- Substituting absolute magnitudes in Solar units, this relates to the luminosity ratio:

$$(g - r) = -2.5 \log_{10} \left(\frac{L_g/L_{\odot,g}}{L_r/L_{\odot,r}} \right) \implies \log_{10} \left(\frac{\mathcal{L}_g}{\mathcal{L}_r} \right) = -\frac{g - r}{2.5}$$

- Since the physical stellar mass M_* of a galaxy is identical in both bands:

$$\frac{(M/L)_r}{(M/L)_g} = \frac{M_*/\mathcal{L}_r}{M_*/\mathcal{L}_g} = \frac{\mathcal{L}_g}{\mathcal{L}_r}$$

- Taking the logarithm of both sides gives the translation formula:

$$\log_{10}(M/L)_r = \log_{10}(M/L)_g - \frac{g - r}{2.5}$$

3. **Step 3: Absolute Magnitude (M_r):** Absolute magnitude represents the galaxy's intrinsic luminosity. We calculate M_r relative to the Solar absolute magnitude in the r -band ($M_{\odot,r} = 4.65$):

$$M_r = M_{\odot,r} - 2.5 \log_{10} \left(\frac{L_r}{L_{\odot,r}} \right)$$

4. By definition of the mass-to-light ratio, the luminosity in Solar units is:

$$\frac{L_r}{L_{\odot,r}} = \frac{M_{\star}}{(M/L)_r}$$

5. Substituting this yields our absolute magnitude formula:

$$M_r = 4.65 - 2.5 \cdot (\log_{10} M_{\star} - \log_{10} (M/L)_r)$$

where $M_{\star}$ is the stellar mass in Solar units.

4. **Step 4: Cosmology and Luminosity Distance ($D_L(z)$):** In a flat Λ CDM cosmology ($H_0 = 70.0$ km/s/Mpc, $\Omega_{m,0} = 0.3$), the comoving distance is integrated:

$$D_C(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_{m,0}(1+z')^3 + \Omega_{\Lambda,0}}}$$

5. The luminosity distance is:

$$D_L(z) = (1+z) \cdot D_C(z)$$

To speed up computations, we pre-compute a grid of (z, D_L) and interpolate using `scipy.interpolate.interp1d`.

Distance Modulus & Apparent Magnitude

- Absolute magnitude M_r is defined at a standard distance of exactly 10 pc. Since flux scales as $f \propto 1/d^2$:

$$\frac{f(D_L)}{f(10 \text{ pc})} = \left(\frac{10 \text{ pc}}{D_L} \right)^2 \implies m_r - M_r = 5 \log_{10} \left(\frac{D_L}{10 \text{ pc}} \right)$$

- To use $D_L(z)$ in Megaparsecs (Mpc), where $1 \text{ Mpc} = 10^6 \text{ pc}$, we convert:

$$\frac{D_L(\text{in pc})}{10 \text{ pc}} = \frac{D_L(\text{in Mpc}) \times 10^6}{10} = D_L(\text{in Mpc}) \times 10^5$$

- Substitute into the distance modulus equation:

$$m_r - M_r = 5 \log_{10} (D_L(\text{Mpc}) \times 10^5) = 5 \log_{10} D_L(\text{Mpc}) + 25$$

- This gives the final apparent magnitude formula:

$$m_r = M_r + 5 \log_{10} D_L(\text{Mpc}) + 25$$

Worked Example: Malmquist Bias in Action

Let's compute apparent magnitudes for a galaxy with $M_{\star} = 10^{10} M_{\odot}$ ($\log_{10} M_{\star} = 10.0$) and color $(g - r) = 0.6$:

- 1. **Mass-to-light ratio:** $\log_{10}(M/L)_g = 1.5(0.6) - 0.7 = 0.2$
- 2. **translation to r-band:** $\log_{10}(M/L)_r = 0.2 - 0.6/2.5 = -0.04$
- 3. **Absolute magnitude:** $M_r = 4.65 - 2.5(10.0 - (-0.04)) = -20.45$

Now let's compare two distances:

- 1. **Close proximity** ($z \approx 0.08$, $D_L = 360$ Mpc):

$$m_r = -20.45 + 5 \log_{10}(360) + 25 = 17.33 \implies \text{Detected } (17.33 < 23.5)$$

- 2. **Distant universe** ($z \approx 0.95$, $D_L = 6200$ Mpc):

$$m_r = -20.45 + 5 \log_{10}(6200) + 25 = 23.51 \implies \text{Masked Out } (23.51 \geq 23.5)$$

This proves why distant dwarf galaxies vanish from our survey!

6. **Step 6: Selection Mask:** We apply the spectroscopic detection filter:

$$\text{detected_mask} = m_r < 23.5$$

Galaxies with $m_r \geq 23.5$ are masked out, transforming our volume-complete catalog into a flux-limited observational catalog.

Computational Consequence

Applying this mask reduces our mock sample size from $\sim 100,000$ to $\sim 20,000$ galaxies, accurately mimicking real spectroscopic selection effects.

Section 2: Supernova Progenitors & Data Ingestion (Task 8)

Supernova Progenitors: Core-Collapse SNe

- **Progenitors:** Massive stars ($M \geq 8M_{\odot}$, spectral classes O and B).
- **Lifespans:** Extremely short ($\tau \leq 50$ Myr), meaning they explode almost immediately after their parent molecular cloud collapses.
- **Host Environment:** Because massive stars die before they can migrate out of their birth environments, Core-Collapse SNe directly trace instantaneous star formation. They reside exclusively in active star-forming galaxies (spirals and starbursts).
- **Data source:** Confirmed observational catalog table from *Sharma et al. 2024*.

Key Stellar Property

If a galaxy is quenched (no active star formation), it cannot host a Core-Collapse Supernova.

Supernova Progenitors: Type Ia SNe

- **Progenitors:** Carbon-Oxygen white dwarfs in binary systems.
- **Explosion Trigger:** Mass transfer from a giant donor star (single-degenerate) or the merger of two white dwarfs (double-degenerate).
- **Delay Time Distribution (DTD):** Unlike massive stars, white dwarfs can take a long time to explode. The delay time τ follows a power-law distribution $\text{DTD}(\tau) \propto \tau^{-\gamma}$ (with $\gamma \approx 1.15$), spanning from ~ 100 Myr to > 10 Gyr.
- **Host Environment:** SNe Ia occur in both active, young star-forming galaxies and old, dead elliptical galaxies. The rate depends on both the total stellar mass (older stars) and the Star Formation Rate (younger stars).
- **Data source:** SDSS-II Supernova Survey catalog (*Lampeitl et al. 2010*).

Supernovae explode within the interstellar medium of their host galaxies, often offset by several kiloparsecs from the galactic nucleus.

- On the sky, this offset corresponds to angular separations of several arcseconds.
- We must associate each supernova coordinate (ra_{SN}, dec_{SN}) with the persistent host galaxy.
- **Spatial Search Box:** We define an angular bounding box of 15 arcseconds (0.0042°):

$$\Delta\alpha = |ra_{\text{host}} - ra_{SN}| \leq 0.0042^\circ$$

$$\Delta\delta = |dec_{\text{host}} - dec_{SN}| \leq 0.0042^\circ$$

- If multiple galaxies fall within this box, we calculate the exact spherical angular separation $\Delta\theta$ and select the closest:

$$\Delta\theta = \sqrt{(\Delta\alpha \cos dec_{SN})^2 + (\Delta\delta)^2}$$

Once we match coordinates, we fetch the host galaxy properties by querying the SDSS DR17 spectroscopic database server.

SDSS Tables Used

- **PhotoObjAll (p):** Contains photometric measurements (coordinates, magnitudes in u, g, r, i, z).
- **SpecObj (s):** Contains spectroscopic measurements (redshift z , spectral class).
- **StellarMassFSPSFit (m):** Contains stellar masses and star formation rates derived by fitting Flexible Stellar Population Synthesis (FSPS) models to the galaxy spectra.

SQL Join Condition

```
PhotoObjAll p JOIN SpecObj s ON p.objid = s.bestObjID JOIN  
StellarMassFSPSFit m ON s.specObjID = m.specObjID
```

- **Web Application Firewall (WAF) Block:** The SDSS spectroscopic server is protected by a firewall. It rejects standard multi-line SQL queries as a security precaution, returning a 403 Forbidden error.
- **Bypass Implementation:** We must programmatically sanitize and flatten our SQL query into a single-line string, stripping out newlines (`\n`) and excess spacing:
 - *Before:* `SELECT ... \n FROM ... \n WHERE ...`
 - *After:* `SELECT ... FROM ... WHERE ...`
- **Batching:** To avoid 503 Gateway Timeouts when querying many coordinates, we split our list of 50 SNe Ia coordinates into batches of 10 and query them sequentially.
- We implement rate-limiting (pause 1s between queries) and connection retry loops.

Section 3: Statistical Testing & Results (Task 9)

Why can't we use standard student's t-tests or ANOVA to compare our supernova hosts with background galaxies?

- **The Assumption Violation:** Parametric tests assume the underlying distributions are Normal (Gaussian).
- Galaxy stellar masses follow a Schechter function (power-law at low masses, exponential cutoff at high masses).
- Galaxy star-formation rates are highly bimodal (star-forming main sequence vs. quenched red sequence).
- **The Solution:** We must use non-parametric tests like the 1D/2D Kolmogorov-Smirnov (K-S) and Anderson-Darling (A-D) tests, which make no assumptions about the shape of the distribution.

Interpreting the Mass-sSFR Plane Results (1/2)

Executing our verification code yields the following results:

Test / Variable	CC-SNe Hosts vs. Background	Type Ia Hosts vs. Background
1D K-S Stellar Mass	$p \approx 0.08$ (Not Significant)	$p \approx 10^{-3}$ (Highly Significant)
1D K-S sSFR	$p \approx 10^{-4}$ (Highly Significant)	$p \approx 0.05$ (Marginal)
2D K-S Joint Plane	$p \approx 10^{-5}$ (Very Significant)	$p \approx 10^{-6}$ (Extremely Significant)

Physical Explanation

- **CC-SNe:** Have masses similar to average galaxies, but have much higher star-formation rates.
- **Type Ia:** Reside in more massive galaxies due to the rate scaling with mass, but have a mixed star-formation profile.
- **2D K-S Power:** By evaluating the joint Mass-sSFR plane, the 2D test captures the correlation structure (SFMS vs. Red Sequence) and reaches high statistical significance.

What We Covered Today

- How telescopes select light (apparent magnitude, flux limit, Malmquist Bias).
- Progenitor physics (Core-Collapse vs. Type Ia environments).
- Coordinate cross-matching using spatial search boxes.
- Querying SDSS and sanitizing queries to bypass firewalls.
- 1D and 2D non-parametric tests on host catalogs.

Session 2 Preview: Next session, we will address class imbalances using SMOTE under strict train-test separation to prevent data leakage, and train our first MLP classifier.